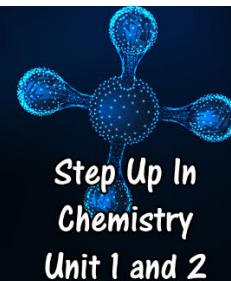


2.1 Compounds and Mixtures

Problems Worksheet



1. Define the following terms:
 - a. Element
 - b. Compound
 - c. Mixture
2. Classify the following as elements, compounds or mixtures.
 - a. Salt
 - b. Iodine salt
 - c. Nitrogen gas
 - d. Compressed air
 - e. White coffee
 - f. Distilled water
 - g. Sulfur

3. Describe the two different types of mixtures.
4. Solder is an alloy often made from tin and silver. The melting point of tin is 232 °C and the melting point of silver is 962 °C. However, there is no fixed melting point for solder. Explain this observation.
5. Describe a procedure that could be used to separate the following mixtures:
 - a. Sand and salt
 - b. Turpentine and water

6. During fermentation to produce alcoholic beverages, methanol is sometimes produced as a by-product of the fermentation, especially when fruit is fermented. Methanol is a toxic compound which can cause blindness and death if it is consumed. The boiling point of methanol is 65 °C while the boiling point of ethanol is 78 °C.
- a. How could the methanol be separated from the mixture?

b. Why is this still not recommended without expertise?

7. Label the diagram of the fractionating column using the words provided.

Crude oil, Kerosene ($C_{10} - C_{16}$), Petrol ($C_5 - C_{10}$), Lubricating oil ($C_{20} - C_{50}$), Fuel oil ($C_{50} - C_{70}$), Naphtha ($C_8 - C_{12}$), Refinery gas ($C_1 - C_4$), Bitumen ($>C_{70}$), Diesel oil ($C_{14} - C_{20}$)

